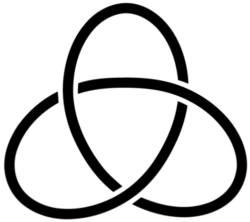
Introduction to Knot Theory

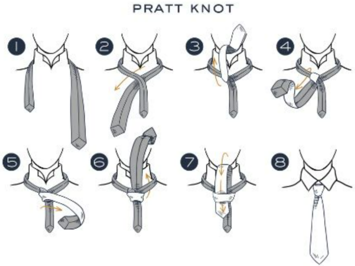
(1) History, Knot Equivalence, Reidemeister Moves



Seoul National University

2026.3.13

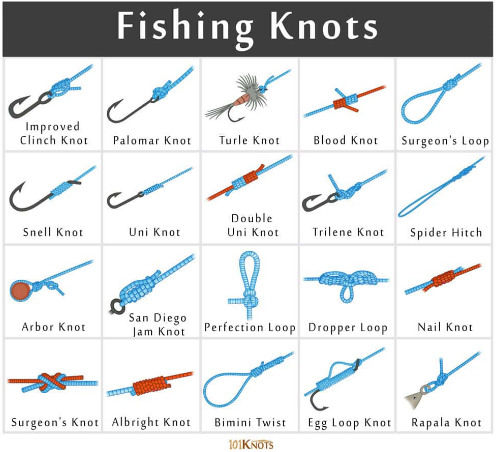
Knots in Everyday Life



Left: Shoelace knot — `https://www.instagram.com/reel/DLPx9CZOVXu/`

Right: Tie knot — `https://www.tie-doctor.co.uk/blogs/news/types-of-tie-knots`

Knots in Crafts and Hobbies



Left: Macrame knot — `https://www.cchobby.com/a-braided-wall-hanging-with-macrame-from-cotton-tube-yarn`

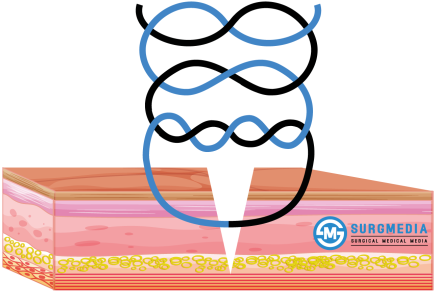
Right: Fishing knots — `https://www.netknots.com/fishing_knots`

Knots in Professional Work

Left: Sailing knots — https://www.slam.com/blogs/slam-stories/different-types-of-sailing-knots-a-practical-guide-for-sea-lovers

Right: Rescue rope knot — https://www.instagram.com/p/DCy5TcgPHuS/

Knots in Science

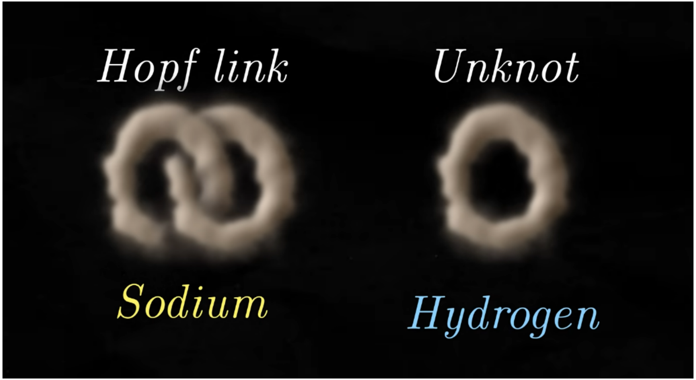


Left: Surgical knot — `https://surgmedia.com/surgical-knot-types-knot-tying-techniques/`

Right: Tangled earphones — `https://abcnews.com/Technology/scientist-untangles-mystery-jumbled-headphones/story?id=24465812`

History of Knot Theory: Ether Model

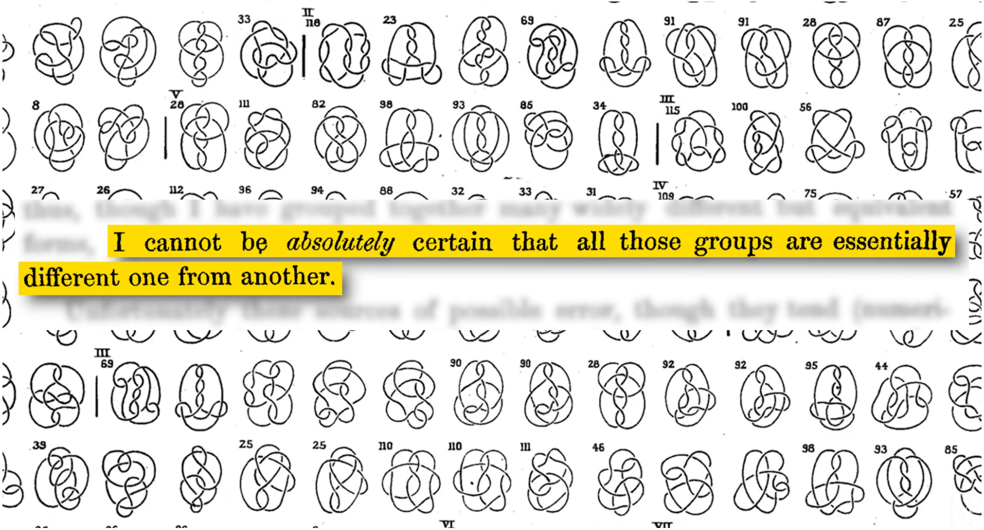
- In 1867, **Lord Kelvin** proposed that atoms were knotted vortices in the ether.
- This idea motivated the first systematic study of knots.
- The physical model was abandoned, but the mathematics survived.



History of Knot Theory: Early Knot Tables

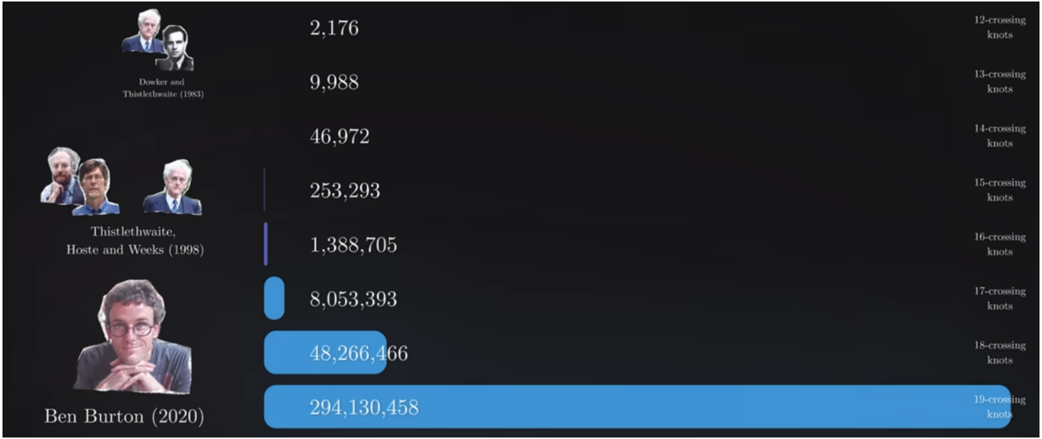
- **Tait** classified knots by crossing number.
- These were among the earliest topological classification tables.

History of Knot Theory: Early Knot Tables

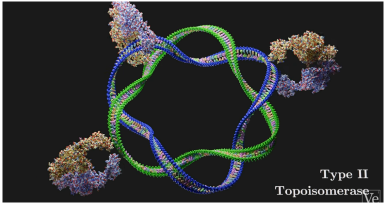


History of Knot Theory: Modern Knot Enumeration

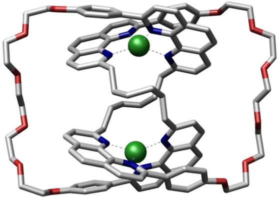
- Modern algorithms allow enumeration of knots with very large crossing numbers.
- For example, Ben Burton (2020) extended knot tables far beyond earlier classifications.



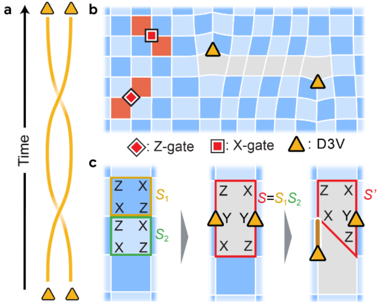
History of Knot Theory: Applications



DNA / protein topology



Molecular knots



Anyon braiding (topological quantum computing)

Definition of a Knot

Mathematical Definition

A **knot** is an embedding

$$K : S^1 \hookrightarrow S^3$$

- S^1 : circle
- S^3 : 3–sphere (usually we work in $\mathbb{R}^3$)
- The curve must be **closed** and **non-self-intersecting**

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Knot vs Link

- **Knot** : one closed loop
- **Link** : several loops

Definition of a Knot

Why Codimension Two?

For embeddings $S^k \hookrightarrow S^n$:

- If $n - k \geq 3$, there is enough room to untangle embeddings.
- Nontrivial knotting first appears in codimension 2.

$$S^1 \hookrightarrow S^3$$

is the simplest nontrivial case.

Ambient Isotopy

Definition

Two knots K_0 and K_1 are said to be **equivalent** if there exists a continuous family of homeomorphisms

$$F_t : S^3 \to S^3 \quad (0 \leq t \leq 1)$$

such that

$$F_0 = \mathrm{id}, \qquad F_1(K_0) = K_1.$$

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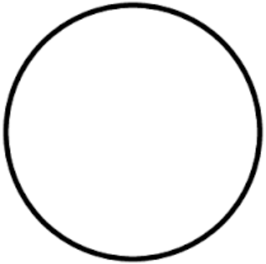
Intuitively, one knot can be continuously deformed into the other without cutting the string or passing strands through each other.

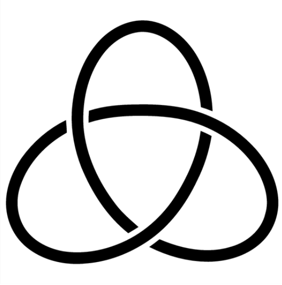
Examples of Knots

Unknot



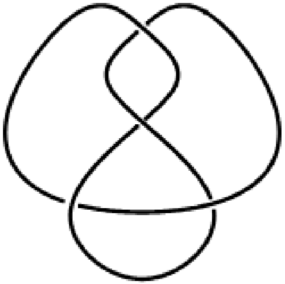
trivial knot

Trefoil Knot



crossing number 3

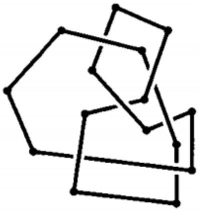
Figure-Eight Knot



crossing number 4

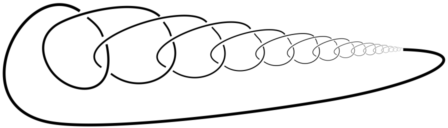
Examples of Knots

Polygonal Knot



piecewise linear (PL) knot

Wild Knot



infinitely complicated local structure

Polygonal Knots

Polygonal Curve

Let $v_0, v_1, \ldots, v_n \in S^3$. A **polygonal curve** is

$$\bigcup_{i=0}^{n-1} [v_i, v_{i+1}],$$

where $[v_i, v_{i+1}]$ is the line segment joining v_i and v_{i+1}.

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If $v_n = v_0$ and the curve has no self-intersections, it is called a **polygonal knot**.

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Equivalent Description

A polygonal knot is a **piecewise linear (PL) embedding**

$$S^1 \hookrightarrow S^3.$$

Tame and Wild Knots

Tame Knot

A knot is called **tame** if it is ambient isotopic to a polygonal knot.

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Wild Knot

A knot that is not tame is called a **wild knot**.

Remark

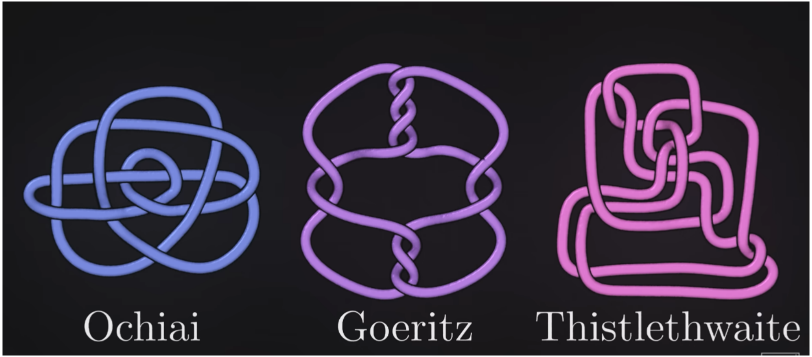
In dimension three the following categories essentially agree

$$\mathrm{Top} \approx \mathrm{PL} \approx C^\infty.$$

Hence every tame knot is isotopic to a smooth knot.

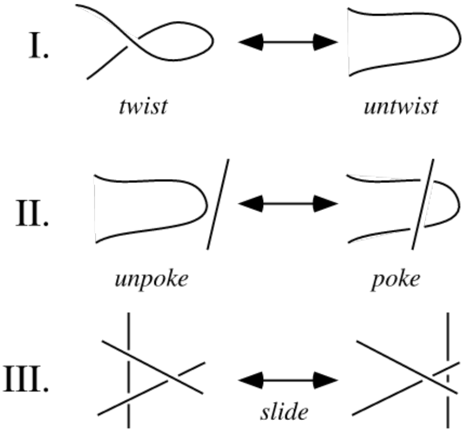
Knot Diagram

A knot diagram is a planar projection of a knot with crossing information.



Different diagrams may represent the **same knot** (here, the unknot).

Reidemeister Moves



Reidemeister Theorem

Two knot diagrams represent the same knot iff they are related by Reidemeister moves.

Reidemeister Moves: Examples

Figure 1.26 Reidemeister moves.

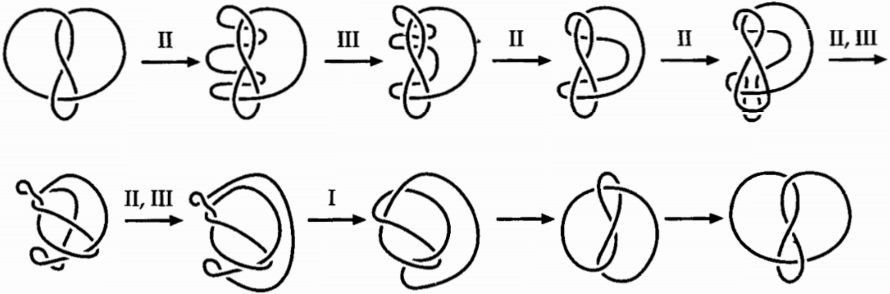


Figure 1.27 The figure-eight knot is equivalent to its mirror image.

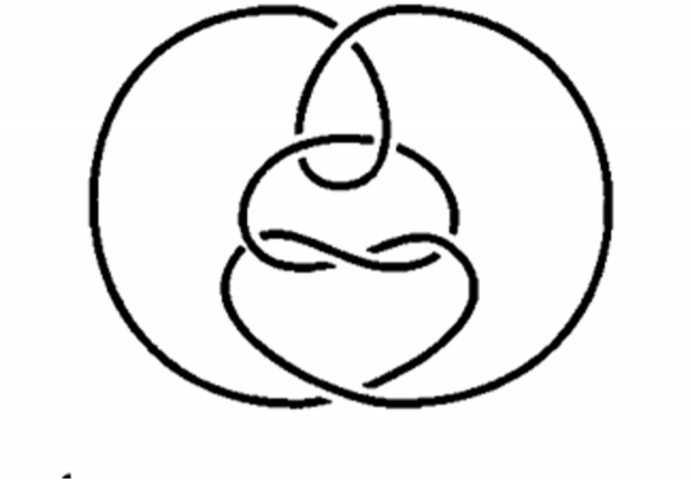
Reidemeister Moves: Exercises

Show that the following two diagrams represent **equivalent knots**.

Show that the following two diagrams represent **equivalent knots**.

Reidemeister Moves: Exercises

Show that the following diagram represents the **unknot**.



The Unknotting Problem

Problem

Given a knot diagram, determine whether it represents the **unknot**.

- A complicated diagram may still represent the unknot.
- We must determine whether it can be simplified to a circle using Reidemeister moves.

Major Results

- **Haken (1961)** Unknot recognition is **decidable**; an algorithm exists.
- **Hass–Lagarias–Pippenger (1999)**

$$\text{Unknot recognition} \in NP$$

- **Kuperberg (2011)**

$$\text{Unknot recognition} \in coNP$$

assuming the **Generalized Riemann Hypothesis**.

The Unknotting Number

Definition

The **unknotting number** $u(K)$ of a knot K is the minimum number of crossing changes required to transform K into the unknot.

- A **crossing change** switches an over-crossing to an under-crossing (or vice versa).
- The unknot has $u(K) = 0$.
- Many nontrivial knots can become the unknot after only a few crossing changes.

Examples

$$u(\text{trefoil}) = 1, \qquad u(\text{figure-eight}) = 1.$$

Complexity Result

Computing the unknotting number is **NP-hard** (Lackenby, 2015).

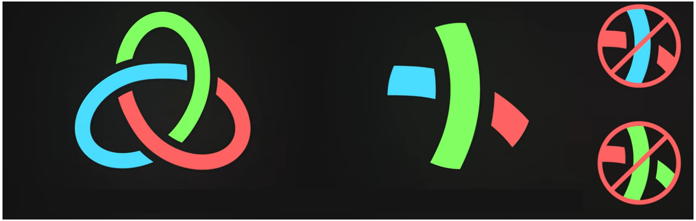
Tricolorability

Definition

A knot diagram is **tricolorable** if

- at least two colors are used,
- at each crossing, the three incident arcs are either all the same color or all different colors.

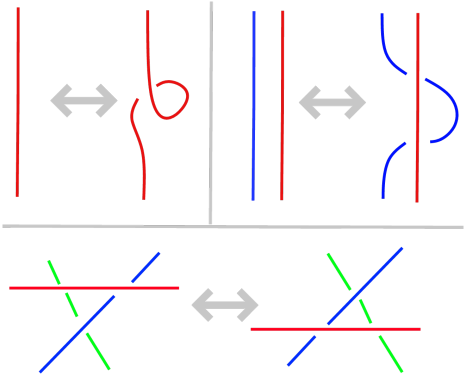
Trefoil is tricolorable, but the unknot is not.



Tricolorability

Proposition

Tricolorability is preserved under Reidemeister moves. Hence it is a **knot invariant**.



Summary

- A **knot** is an embedding
$$S^1 \hookrightarrow S^3$$
- Two knots are considered equivalent via **ambient isotopy**.
- **Reidemeister moves** characterize equivalence of knot diagrams.
- Knot invariants such as **tricolorability** help distinguish knots.
- The **unknot recognition problem** asks whether a diagram represents the unknot.
- The **unknotting number** measures how many crossing changes are needed to obtain the unknot.

References

References

- Veritasium, *The Surprisingly Beautiful Math of Knots*
- C. C. Adams, *The Knot Book*
- D. Rolfsen, *Knots and Links*

Image Sources

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- Molecular Knot — Wikipedia
- Knot Diagram — Wikipedia
- Reidemeister Moves — Wolfram MathWorld
- Tricolorability — `https://jorisbukala.com/posts/knot-theory/`
- Reidemeister Moves Example, Exercises — Adams, *The Knot Book*
- Anyon Braiding — Nature (2023)
  `https://www.nature.com/articles/s41586-023-05954-4`